



Personal Website:
<https://frankc03.github.io/>

YIFEI CUI

Phone: 86-13855116835 | Email: FrankC03@163.com
Address: No.2006, Xiyuan Ave, West Hi-Tech Zone, 611731, China

EDUCATION

University of Electronic Science and Technology of China (UESTC)	09/2021-07/2025
Glasgow College, jointly held with the University of Glasgow	
- Major: Electronic Information Engineering, BEng - GPA: 3.91/4.0, Weighted Average: 89.87/100, Ranking: 4/247 (Top 1.6%)	
University of Glasgow	09/2021-07/2025
- Program: Electronics and Electrical Engineering, BEng - Expected: First-class Degree with Distinction	
Self-preparatory Courses - MIT Graduate Courses for Integrated Circuits Design	05/2023-07/2023
Supervisor: Prof. Cheng Wang (<i>MIT EECS PhD</i>)	
6.374 - Analysis and Design of Digital Integrated Circuits 6.775 - CMOS Analog and Mixed-Signal Circuit Design 6.776 - High-speed Communication Circuits	

AWARDS AND HONORS

National Scholarship (Top 4%)	11/2024
First-class Scholarship for Outstanding Students of UESTC	10/2022&2023&2024
Academic Performance Scholarship (Top 5%)	11/2022&2023&2024
Finalist Prize in the Interdisciplinary Contest In Modeling (Top 2%)	05/2023
Second Prize in the Mathematical Modeling Contest of UESTC	11/2022
Watt Innovative Talent Scholarship	11/2022
UESTC Outstanding Social Practice Individual	12/2022

RESEARCH EXPERIENCE

Neural Network-based Surface Decoder for the Scalable Quantum Error Correction	08/2023-Present
Role: Undergraduate Research Assistant	
Supervisor: Prof. Cheng Wang	
Project Focus: Developing a scalable neural network-based surface decoder for quantum error correction to address the inefficiencies of traditional decoders, such as Minimum Weight Perfect Matching (MWPM), in large-scale quantum systems	
Key Responsibilities:	
Noise Modeling: Designed a quantum noise model covering measurement errors, data qubit idling, syndrome extraction errors, and qubit leakage Model Verification: Compared model accuracy with standard error models using error pair correlation matrices from IBM experiments and model simulations Neural Network Decoder: Initiated the design of a machine learning-based decoder aiming at improving speed and accuracy over MWPM (In Progress)	

PROJECT EXPERIENCES

Intelligent Rover Design	02/2024-06/2024
Role: Leader of a Six-person Team	
- Completed main program design using STM32CubeIDE, implemented tasks via polling and external environment interrupt - Realized visual algorithm with OpenMV and its IDE, adopted H-bridge to design the driver module PCB independently, and finished physical verification - Used STM32F446RE, McNamum wheel, OpenMV Cam, ultrasonic module, DCDC power module, lithium battery and self-designed driven module to achieve the hardware welding of the rover, and finally ensure the functions, including line patrol, direction sign and traffic light recognition, pedestrian and obstacle avoidance, wireless switch parking lever, with over 90% success rate in 5 out of 6 tasks	

Digital CPU Design

02/2023-07/2023

- Datapath Design and Controller Design: Composed a specification for basic CPU, designed ALU component; Understood the STG for the control unit, designed state transition component; Coded, synthesized and conducted verification for datapath component and controller component
- CPU Integration: Combined the datapath and the controller, coded, synthesized and conducted simulation for the CPU
- CPU Verification and Design Improvement: Downloaded the bitstream to FPGA & conducted verification, implemented accumulator function, enhanced CPU feature by enlarging the number of registers and the word-length

Folded Common-source Common-gate Amplifier Design

06/2023-07/2023

- Employed 2.5V devices, under the TSMC 65nm GP process in Cadence Virtuoso to design amplifiers, with requirements: Voltage Gain: 40dB, Bandwidth: 50MHz, Load Capacitance: 10pF, Supply Voltage: 2.0V
- Calculated transconductance, intrinsic gain and current cutoff frequency to complete the Schematic simulation
- Drew layout and performed DRC, LVS and PEX, and finished layout simulation, met all requirements

Weather Clock Embedded System Design

02/2022-06/2022

- Designed the embedded program on Mbed platform, completed the hardware connection via the combination of STM32L432KC single chip microcomputer, esp8266 WIFI module, and OLED display
- Obtained weather time data through the network API, and realized real-time queries and alarm clock setting, and querying the weather of the day and the future

National University of Singapore, NUS, Winter Program

01/2022-02/2022

Artificial Intelligence and Machine Learning – Computer Vision

- Learned the principles and background of neural networks
- Realized traffic sign recognition using the scikit-learn library and convolutional neural network in the python platform

INTERNSHIP

Yangtze Delta Region Institute (Huzhou), UESTC

06/2022-07/2022

Post: Research Outreach Assistant

- Collected and analyzed experimental data on the performance of transmitter and receiver in a spatial millimeter-wave wireless power transfer system, achieving an energy transmission distance of up to 50 meters at 35GHZ.
- Created slides to promote the performance of the system to researchers from other institutes, guided researchers on a tour of the laboratory

EXTRACURRICULAR ACTIVITIES

Volunteer Activities

Community support – Children's Innovation Experience Courses in the city Sci-Tech Museum

12/2022

- Introduced types of renewable energy and guided children in building a simple hand-crank generator to light up an LED
- All children learned principles of sustainable power generation, and successfully lit up the LED

Campus Environmental Protection Publicity

09/2022

- Organize an eco-friendly trivia competition and a litter-picking contest
- Deepened understanding of environmental knowledge, such as waste sorting, and helped cultivate good habits of not littering

School Anniversary Video Shooting

09/2021

- Filmed students performing Scottish dance, highlighting the unique features of Glasgow College
- The filmed clips were compiled into a campus promotional video and received positive response

Sports Competitions

Glasgow College Basketball Class League (Third Place)

05/2023

UESTC, Freshmen Cup Table Tennis Competition (Last 16)

11/2021

Glasgow College Freshmen Cup football participation (Last 8)

09/2021

SKILLS

Professional Tools: Cadence Virtuoso, Xilinx Vivado, HFSS, PSpice, Keil5, LaTeX, MATLAB

Programming Language: C, C++, Python

Others: IELTS- Overall 7.5 (Listening: 9, Reading: 8, Writing: 6.5, Speaking: 7)